FACULTY OF COMPUTING
UTM Johor Bahru

FACULTY OF COMPUTING

SEMESTER 2 / 2023-2024

SECV2113 SECTION 02

HUMAN-COMPUTER INTERACTION

ASSIGNMENT 2
HE JOURNEY MAPPING

Lecturer : Dr Nur Zuraifah Syazrah binti Othman

Student Name	Matric No
Chew Chiu Xian	A23CS0061
Lau Yan Kai	A23CS0098
Danish Afiq bin Roshisham	B22EC0004
Dheshieghan A/L Saravana Moorthy	A23CS0072
Pravinraj A/L Sivabathi	A23CS0171

Introduction

In this assignment 2 of Human-Computer Interaction course, we will be conducting heuristics evaluation of the ShopEase prototype, a fashion shopping application designed by group of COSMOS. This evaluation is a crucial step in the overall project development process. We will be applying Jakob Nielsen's 10 heuristics to systematically assess the user interfaces designed by group of COSMOS and rate them based on the observed severity level of any issue, to enhance the overall user experience of the application.

The 10 Heuristics and the Severity Rating

Heuristics	Severity Rating (1-low, 4-high)
1. Visibility of system status. 2. Match between system and real world. 3. User control and freedom. 4. Consistency and standards. 5. Error prevention. 6. Recognition rather than recall. 7. Flexibility and efficiency of use. 8. Aesthetic and minimalist design. 9. Help users recognize, diagnose, recover from errors. 10. Help and documentation.	1. **Cosmetic issue** – affects the appearance and should be fixed only if time permits 2. **Minor issue** – hinders the user's ability to navigate and should be fixed when possible 3. **Major issue** – Frustrates or confuses users and requires repair as soon as possible 4. **Catastrophic issue** – prohibits users from performing their given task and requires an immediate modification

By evaluating the ShopEase prototype using Jakob Nielsen's 10 heuristics, we can identify any usability issues and provide feedback to the COSMOS team. This feedback will enable them to make the necessary revisions to enhance the design and improve the design according to societal expectations and user needs. It will helps attract a larger user base upon launch.

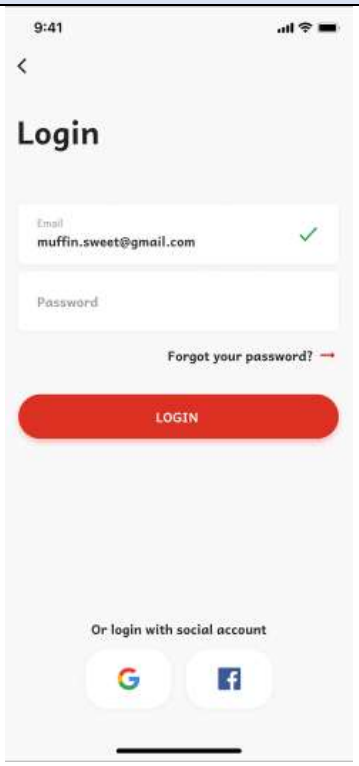
Our evaluation method involves collaborative group discussions amongst all team members. Each member will first gain an understanding of the heuristic evaluation process. Together, we will test the prototypes based on the tasks given by applying Jakob Nielsen's 10 heuristic rules to critically identify and document any usability violations in the application.

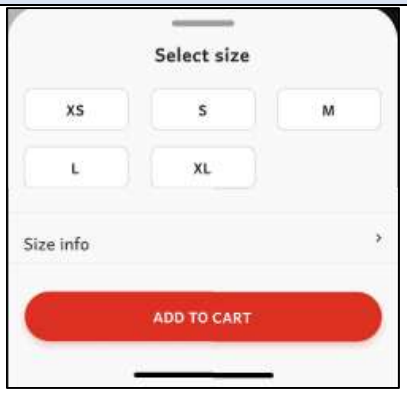
Persona
Alif Satar, an 18-year-old high school graduate who's new to using different online shopping websites.
Scenario
As Alif continues his search for the perfect college bag, he decides to explore beyond Lazada and try out Shopee, prompted by friends' claims of better deals. Although navigating Shopee proves initially perplexing for him as a first-timer, Alif is intrigued by its voice assistant and narration features, which ultimately aid him in finding his way around the platform. To his surprise, he discovers multiple listings for the same bag on Shopee, each with slight variations in price and shipping terms. Alif meticulously compares these details, weighing any additional discounts or promotions offered. After careful consideration, he notices that while Shopee's initial price is slightly lower, its shipping fees are higher than Lazada's free shipping offer. After pondering the pros and cons, Alif concludes that Lazada's overall cost-effectiveness and convenience outweigh the marginally lower price on Shopee. Confident in his decision after thorough comparison, he proceeds to make his purchase on Lazada, content with his informed choice.

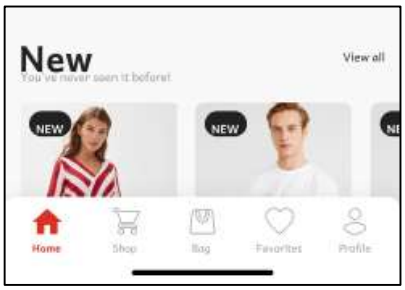
Given tasks

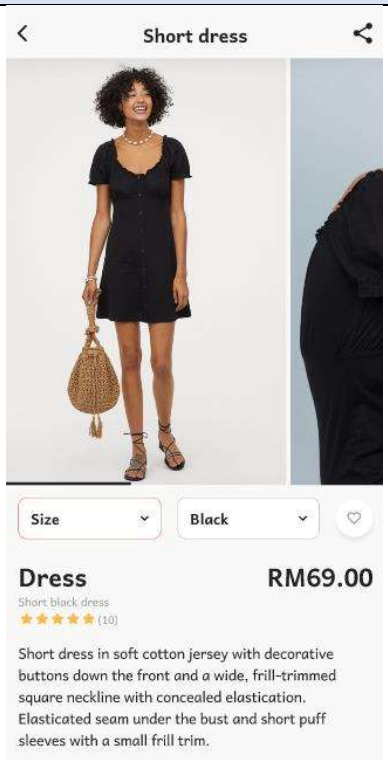
1. Search a beauty product using voice assistant function
2. Use narration feature to review product description
3. Use the Help popup feature to assist in adding product to cart

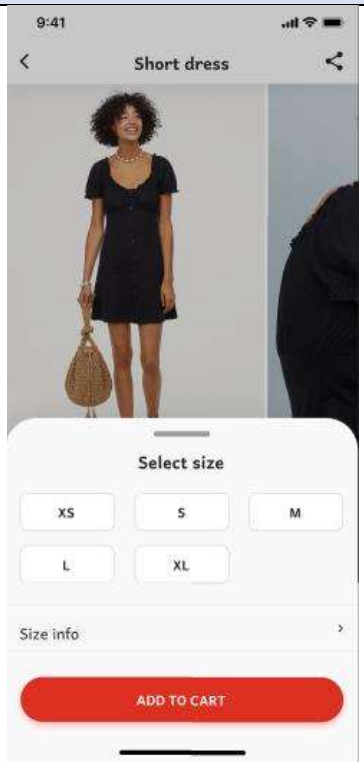
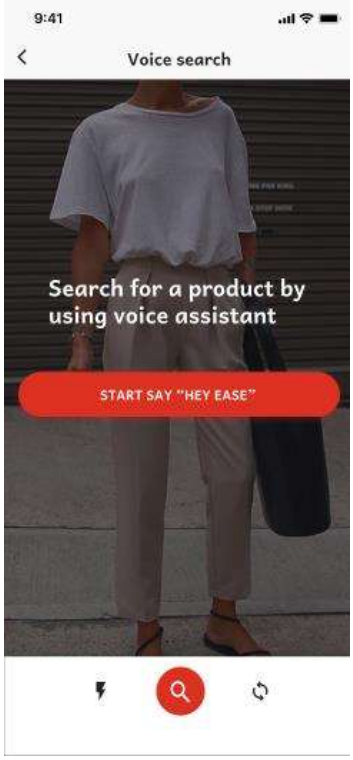
Heuristic Evaluation (HE) Table

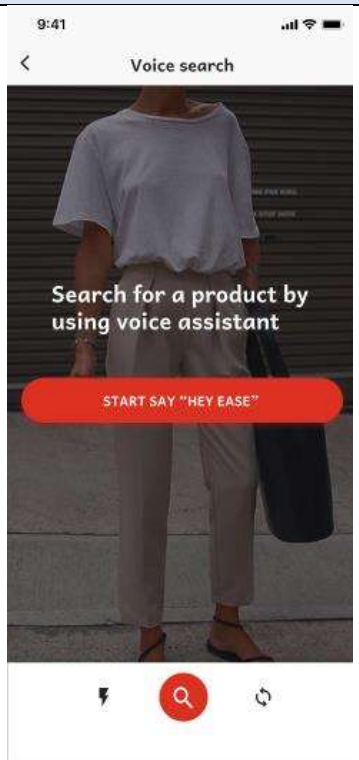
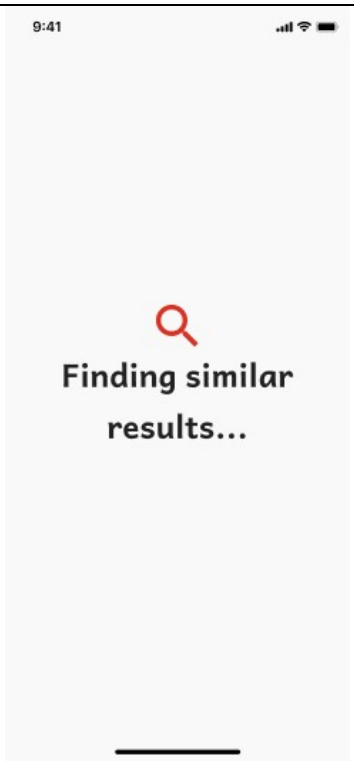
No	Prototype Image	Identified Issue	Heuristic and Severity
1		There is no loading, error or successful message for the user on sign up or login page. It may cause confusion and repeated actions of the process.	**H1** Visibility of System Status **S3** Major issue
2		The guidance page for adding items into cart is cluttered. It overwhelmed the user who are first time of using this application. It will cause confusion and frustration for the new users.	**H8** Aesthetic and minimalist design **S1** Cosmetic issue

<table>
<tr><th>No</th><th>Prototype Image</th><th>Identified Issue</th><th>Heuristic and Severity</th></tr>
<tr><td>3</td><td> 


</td><td>The process is added to cart and state “Add to Cart”, however in main menu, the cart is called bag and the icon used is also bag.</td><td><table><tr><td>H4</td><td>Consistency and Standard</td></tr><tr><td>S2</td><td>Minor issue</td></tr></table></td></tr>
<tr><td>4</td><td>
</td><td>No feedback/messages after adding an item to the cart. It will cause the user to question if the action is successful and cause frustration.</td><td><table><tr><td>H1</td><td>Visibility of system status</td></tr><tr><td>S3</td><td>Major issue</td></tr></table></td></tr>
</table>

<table>
<tr><th>No</th><th>Prototype Image</th><th>Identified Issue</th><th>Heuristic and Severity</th></tr>
<tr><td>5</td><td> 
</td><td>No guidance on the narration feature. The user could not understand the narration feature without proper guidance or icon indicated.</td><td><table>
<tr><td>H10</td><td>Help and documentation</td></tr>
<tr><td>S4</td><td>Catastrophic issue</td></tr>
</table></td></tr>
</table>

<table>
<tr><th>No</th><th>Prototype Image</th><th>Identified Issue</th><th>Heuristic and Severity</th></tr>
<tr><td>6</td><td> 
</td><td>User unable to undo their actions, navigate freely, after they add their item into cart, such as modify the item in their cart. It reduces the flexibility of the shopping process.</td><td><table><tr><td>H3</td><td>User control and freedom</td></tr><tr><td>S3</td><td>Major issue</td></tr></table></td></tr>
<tr><td>7</td><td> 
</td><td>There could be unnecessary complexity like “START SAY HEARSE.” It can confuse users and complicate the interaction unnecessarily.</td><td><table><tr><td>H8</td><td>Aesthetic and minimalist design</td></tr><tr><td>S2</td><td>Minor issue</td></tr></table></td></tr>
</table>

No	Prototype Image	Identified Issue	Heuristic and Severity
8		Voice assistant but the control bar is taking picture. Inconsistent UI can mislead user and cause the user about the functionality of the button.	**H4** Consistency and Standard **S3** Major issue
9		The app lack of error prevention mechanism for voice assistant. There is no confirmation for the voice assistant. For example, the user hasn't finished speaking, but the app has started to search.	**H5** Error prevention **S4** Catastrophic issue

User Journey Map

User: Alif Satar
an 18-year-old high school graduate who's new to using different online shopping websites.

Scenario: Prompted by friends' claims of better deals, Alif explores Shopee for a college bag with cheaper price, but he is facing difficulties doing so due to lack of experience in online shopping.

User Goal: Buy a new bag for college from online shopping platforms.

- TASK 1 -	- TASK 2 -	- TASK 3 -
Search a beauty product using voice assistant function	Use narration feature to review product description	Use the Help popup feature to assist in adding product to cart
1. **Open ShopEase app** 2. **Sign in** 3. **Click Shop** 4. **Activate the Voice Assistant** 5. **Start "Say Hey Ease"** "I love how I can immediately find the thing that I wanted to buy using voice assistant!"	6. **Back to home page** 7. **Click Shop** 8. **Select item category** 9. **Select product** 10. **Use narration feature** "Oh no! I can't find the narration feature in the app! There is no icon/guidance on that!"	11. **Back to home page** 12. **Click Help icon** 13. **Click "Yes"** 14. **Click "Next" to continue guide** "It's good to have guidance! However, the guidance is messy and hard to understand."
Recommendation 1. Simplify the UI to improve the user experience and reduce confusion ***(Issue 7)*** 2. Modify the control bar to match the voice assistant features ***(Issue 8)*** 3. Add start and stop icon to show when the voice recording start and stop ***(Issue 9)***	**Recommendation** 1. Show a narration button at the description field or provide narrator guidance ***(Issue 5)***	**Recommendation** 1. Make the guidance simple and easy to understand by simplifying the UI to prevent confusion and frustration. ***(Issue 2)***